Project Review: Optimizing Email Marketing Campaign Using Machine Learning

Overview

The overall goal of this project was to maximize the efficiency of an email-marketing campaign using historical data and by relying on machine learning to maximize Click-Through Rate (CTR). The approach was to help find trends in user behavior and present practical solutions with predictive recommendations.

Project execution

I started this project by performing intensive data cleaning and preprocessing. I dealt with missing values, scaled numerical variables, had to make one sample data as mentioned in Repository in GITHUB, and email types and days of the week. This was an important step towards making the data fit in machine learning algorithms.

I did Exploratory Data Analysis (EDA) after preparing the data. In this, it was associated with charts of major performance metrics like open rates and click rates of various types of emails. As an example, I generated a pie chart to demonstrate the distribution of the open rate and a bar chart that pinpoints the click rate of a specific type of email. These visualizations demonstrated two things, first, transactional emails were the most engaged, and, second, some days and times performed better than others.

New Training model

I use Random Forest Regressor to train a model because it is very robust and takes non-linear patterns of data. I also tried other models such as

Linear Regression and Decision Tree but I realized Random Forest produced better accuracy. The model was tested and trained to forecast CTR depending on the input features

Trained model was able to better predict than baseline models by a margin of about 8 to 15 percent. I will suggest some measures based on the analysis of bettering the performance of emails like time of sending and sending strategy based on the nature of emails.

Reasons to using my methods.

The main reason I decided to use Pandas and NumPy data manipulation libraries is because of their efficient and readable syntax. In case of visualizations, I preferred Matplotlib and Seaborn because I could create understandable and professional-appearing plots. Random Forest Regressor was selected because it is not prone to the overfitting of the complex relations.

Result from new model.

The last model gave a valid method of forecasting the click-through rates on the email messages, so that the marketing team can plan working on the evidence-based ideas. The project proved why data science integration with business requirements is important in improving the effectiveness of campaigns. The visualization and predictive model provided practical recommendations in streamlining future campaigns.

Learning for me in this project.

This project allowed me to learn by doing the whole process of machine learning, i.e. data cleaning, model training and model evaluation. I came to know the significance of the proper preprocessing in the accuracy of models and how to ease complex information using visualizations to make it understandable by the stakeholders. I gained new skills in tools like Github, I learned to compare various algorithms and to present technical information in layman terms as well.This project not only strengthened my technical foundation but also helped me understand the importance of aligning analytics with real-world business goals.